

| Question | Answer                                                                                                                                       | Marks     |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 3(a)     | (thermal) energy per unit mass (to cause state change)                                                                                       | <b>B1</b> |
|          | (thermal) energy to change state at constant temperature                                                                                     | <b>B1</b> |
| 3(b)     | (for vaporisation):<br>involves greater change in volume (of substance)<br><b>or</b><br>involves greater increase in separation of molecules | <b>B1</b> |
|          | more work has to be done by molecules (to separate)<br><b>or</b><br>greater increase in potential energy of molecules                        | <b>M1</b> |
|          | kinetic energy of molecules unchanged, so more thermal energy needed                                                                         | <b>A1</b> |
| 3(c)     | $Q = mc\Delta\theta$ <b>and</b> $Q = mL$                                                                                                     | <b>C1</b> |
|          | $\Delta\theta$ for the water = 26.4 – 10.3                                                                                                   | <b>C1</b> |
|          | $(37.0 \times L) + (37.0 \times 4.18 \times 10.3) = (208 \times 4.18 \times 16.1)$                                                           | <b>C1</b> |
|          | $L = 335 \text{ J g}^{-1}$                                                                                                                   | <b>A1</b> |

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